Effect of Ethanolic Leaf Extract of *Telfairia Occidentalis* on Lipid Profile in Experimentally Induced Chronic Stressed Rat

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ABSTRACT

This study aimed to investigate the effect of the ethanol leaf extract of *Telfairia occidentalis* (*T. occidentalis*) on lipid profile in experimentally induced chronic stressed rats. In a bid to achieve this objectives, 30 male Wistar rats were randomly assigned to six groups: negative control, positive control (untreated chronic stress), standard drug treated (sertraline 25mg/kg), and three test groups treated with 100, 200, or 400 mg/kg of *T. occidentalis* extract. Chronic stress was induced through immobility/retrain stress 2 hours daily for 21 days. By day 22, behavioural studies were carried out using elevated plus maize and open field boxes and blood samples obtained for lipid profile assessment and the data analysed for significant different at $p<0.05$. Behavioral studies revealed low and high doses of *T. occidentalis* extracts to mitigated anxiety-like behaviors induced by chronic stress. There was a non-significant different in serum total cholesterol, triglyceride, High density lipoprotein (HDL) and low density lipoprotein levels between groups, however, there were relatively lower in the *T. occidentalis* treated groups compared with the positive control; except for serum HDL level. This findings suggest that ethanol leaf extract of *T. occidentalis* have potential therapeutic benefit in managing chronic stress induced anxiety, but it effect in liver lipid is indecisive and hence the need for further studies.

Keywords: *Telfairia occidentalis*, Chronic stress, Behavoural studies, Lipid profile, Rats.

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1. INTRODUCTION

Chronic stress has been associated with chronic illness such as obesity and depression and contributes to worse prognosis, accelerated disease progression, and poorer survival in many diseases including lung pathologies [1], cardiovascular disease [2], and cancer [3]. In addition to altered mood and depressive or anxious behavior, chronic stress has been linked with increased inflammation [4]; [5]; [6]; [7], hematopoietic stem cell activation [1], and functional impacts on immune populations [1]; [2] and on erythropoiesis [8].

Given the significant impact of chronic stress on patient morbidity and mortality, it is important to understand the physiological impact of chronic stress, and to identify ways in which the stress response may contribute to exacerbation of pathologies. Elucidating hematologic changes early after stress may provide insight into mechanisms of disease exacerbation and may allow for early detection of stress-related symptomology and pathology. The use of plants in traditional medicine referred to as herbalism or botanical medicine [9] falls outside the mainstream of the Western or Orthodox medicine. It has been estimated that about 75% of the world's population (mainly in the developing countries) rely on traditional medicine as their primary form of health care [10].

The use of traditional medicine in the treatment and management of diseases in Africa cannot fade away and this could be attributed to the socio-cultural, socioeconomic, lack of basic health care and qualified personnel. Our interest of concern is

the *Telfairia occidentalis (T.occidentalis)* which is commonly called fluted pumpkin that is mostly present tropical vine, grown in West Africa and central Africal, expecially in Nigeria, Benin and Cameroon [11]. The medicinal properties of *T. occidentalis* suggest potential therapeutic benefits, and some studies suggest that it have potential health benefits, but more research is needed to extablished its efficacy and safety, hence the need to investigate the effect of ethanolic extract of *T. occidentalis* on liver lipid under chronic stress. Although several medicinal uses of *T. occidentals* in traditional medicine have been documented but many of these claims are yet to be validated by scientific research [12] and thus the setback in ethno medicine and pharmacology. The aim of this study is to investigate the effect of ethanol leaf extract of *T. occidentalis* on lipid profile of experimentally induced chronically stressed Wistar rats.

2. MATERIALS AND METHODS

2.1. Plant materials: The leave of *T. occidentalis* was obtained from a local market in Ekpoma town in Esan West Local Government Area of Edo State, Nigeria and was identified at Forest Research Institute of Nigeria, Ibadan, Nigeria. The plant material was also registered and deposited at the University hebarium of Ambrose Alli University, Ekpoma, Edo State, Nigeria.

2.2. Animals: 30 Male Wistar rats with average weight of 120 -150g was obtained from the animal house of the Faculty of Basic Medical Sciences, Ambrose Alli University, Ekpoma. The rats were housed at room temperature on a 12 hour dark-light cycle and acclimatized for a 14-day period with *ad libitum* access to water and food throughout the period of experiments

2.3. Chemicals, reagents and equipment: All chemicals used were of analytical grade and obtained from known chemical stores or manufacturers. Equipment and facilities used were those available within the Department of Physiology, Ambrose Alli University, Ekpoma, Edo State.

2.4. Extraction of plant material: The ethanol extract of *T. occidentalis* leaf was prepared using the method of Majekodunmi and Nubani [11], with few modifications and the yield was calculated following the equation below. Calculation- Yield (g/100 g) = (W1 × 100)/W2

Where: W1 is the weight of the *extract* residue obtained after ethanol extraction and drying W2 is the weight of dry *T. occidentalis.*

2.5. Animal grouping: The grouping of rats are as followed: Group 1: Negative Control (No exposure, no treatment). Group 2: Positive control (exposed to chronic stress without treatment). Group 3: standard drugs treated; exposed to chronic stress and treated with Sertraline (25mg/kg). Group 4, 5 and 6: Test groups exposed to chronic stress and treated with low dose (100mg/kg), mid dose (200mg/kg) and high dose (400mg/kg) of *T. occidentalis* etract respectively.

Using orogastric tube, the experimental animals were administered *T. occidentalis* extract orally at different doses for 21 days.

2.6. Experimental procedure and sample collection: After acclimatization, rats were received corresponding treatments and 45 minutes after, they were expose to immobilization/restraint stress by placing each animals in an exact size tube for 2 hours daily. At the end of the 21 day exposure, behavioural/ stress assessment was done using the Elevated Plus maze test as previously reported by the method of [12] with few modifications. The behavioural parameters determined comprised of traditional spatio-temporal, locomotion, and ethological measures as documented in [13-15]. The traditional spatio temporal measures include (a) Time on open arms and (b) Frequency of open arm entries. Locomotion and maintenance measures include rearing frequency and duration [16]. The ethiological measures include stretched attend posture, head dips [17] and risk assessment [14].

2.7. Lipid profile analysis: At the end of behavioural assessment, blood samples were obtained into plain (3 ml) bottles for liver profile parameters estimation. The blood samples were centrifuged to obtain serum and using commercially available kits, these were estimated following standard laboratory procedures and inline with the step by step procedure in the manual of kits.

2.8. Statistical analysis: The data were analysed for statistical significance using one-way ANOVA followed by LSD test at $p<0.05$ using the graphpad prism. The results were expressed as mean ± SEM and presented using suitable tables.

3. RESULTS

Table 1 compares behavious assessment in elevated plus maze apparatus. Time spent in the open arm was significantly lower in the positive control compared to the negative control and was significantly higher in the *T. occidentalis* extract treated compared to the positive control. On the other hand, time spent inactive was significant lower in the negative control compared to the positive control and was significantly lower in the low and mid doses of *T. occidentalis* extract treated groups compared to the positive control.

In table 2 compared behavioural assessment in the open field box. Immobility time in the open field was significantly higher in the positive control compared to the negative control and was significantly lower in the low and high doses of *T. occidentalis* extract treated groups compared to the positive control. There was no significant different in the distance travelled by the rats between groups, however, it was higher in the low dose, indifferent in the mid dose and lower in the high dose of *T. occidentalis* extract treated groups compared to the positive control.

In table 3 compares lipid profile of stress exposed rats and treatment with *T. occidentalis*. Serum total cholesterol, triglyceride, high density lipoprotein (HDL) and low density lipoprotein (LDL) were not significantly altered between groups, however, except for HDL, other serum lipids were non-significantly decreased with *T. occidentalis* treatments compared to the positive control group.

Table 1. Stress assessment of rat in elevated plus maze apparatus.

	Negative	Positive	Standard	Low dose	Mild dose	High dose
Time spent in the open arm (secs)	28.87± 5.15	15.00± 1.53*	13.70± 1.76 [a]	39.30± 2.33[a]	53.70± 19.60 [a]	61.30± 4.67 [a]
Time spent inactive (secs)	60.50± 8.63	139.00± 3.53*	99.70± 25.60	66.30± 8.67 [a]	51.30± 21.90[a]	132.00± 15.90*

*= significant compared with control, [a] = significant compared with positive control.

Table 2. Behavioral assessment of rats in open field study.

	Negative	Positive	Standard	Low dose	Mild dose	High dose
Immobility (Sec)	79.30± 3.480	111.00± 5.36*	108.00± 4.36*	14.30± 2.33* [a]	117.00± 1.20*	72.30± 6.49 [a]
Distance traveled(cm)	8.00± 1.67	8.97.± 1.53	6.20± 2.94	12.80± 1.36	8.80± 0.79	4.23± 0.85

*= significant compared with control, [a] = significant compared with positive control.

Table 3. Lipid profile after chronic stress exposure and *T. occidentalis* treatment in rat.

	Negative	Positive	Standard	Low dose	Mild dose	High dose
LDL	25.33±0.57	28.87±2.48	26.87±2.23	25.13±1.27	24.73±1.61	25.80±03.00
Trig	46.66±4.37	45.66±5.23	39.00±2.31	47.33±5.23	43.00±2.08	42.66±3.84
CHOL	77.67±4.84	77.33±4.33	69.33±1.76	73.66±5.54	75.00±2.64	73.33±3.17
HDL	43.00±3.00	39.33±4.84	34.66±3.28	39.33±5.20	41.66±3.84	39.50±1.43

No significant different compared with control and positive control.

4. DISCUSSION

The study assessed the effects of *T. occidentalis* extract on chronic immobility stress induced behavioural and lipid profile alterations in rats. The findings on lipid profile were not significantly different between groups. Although *T. occidentalis* extract treatment were most likely to cause a decreased in serum total cholesterol, triglyceride and LDL but increased serum HDL. Studies have reported the different leaf extract of *T. occidentalis* to lowered cholesterol levels [18-20] and others documenting increase in serum cholesterol levels but a decrease in other serum parameters [21]. There are some studies which reported *T. occidentalis* extract may be beneficial in managing hyperlipidemia by significantly reducing serum triglyceride and highlighting its potential as a dietary supplement [18, 19, 22]. Other studies have documented *T. occidentalis* extract may be a useful therapy for hypercholesterolemia, as the extract significantly increased in HDL levels alongside reductions in total cholesterol [13, 20, 25].

The behavioral studies conducted using the elevated plus maze apparatus provided further insights into the impact of chronic stress and the potential protective effects of *T. occidentalis* extract. The present study showed that chronic immobility stress induced anxiety-like behaviors such as reduced time spent in the open arm and distance travelled but increased duration of immobility and inactivity. These findings align with previous studies demonstrating the detrimental effects of chronic stress on anxiety-like behaviors and overall well-being [24, 25]. Interestingly, treatment with the low and high doses of *T. occidentalis* extract appeared to mitigate some of the anxiety-like behaviors observed in the chronically stressed rats. These findings suggest a potential anxiolytic effect of the *T. occidentalis* extract.

CONCLUSION

Overall, the findings from this study suggest that the ethanol leaf extract of *T. occidentalis* have potential therapeutic benefits in managing chronic stress-induced anxiety-like behaviors but with inconclusive findings on lipid profile. However, it is crucial to note that the observed effects appear to be dose-dependent, with lower doses exhibiting more favorable outcomes compared to the highest dose tested.

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